Harry Wu

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Full-Stack Software Developer - AI Applications

Senior Full-Stack Software Developer with 10+ years of experience delivering customer-facing web applications, enterprise commerce platforms, and microservices, with a growing focus on AI application development. Combines strong expertise in React, TypeScript, Node.js, PHP/Laravel, and AWS with hands-on experience in Python/FastAPI, RAG-powered assistants, and multi-agent workflows using LangChain and LangGraph. Hands-on technical leader experienced in system architecture, performance optimization, production reliability, and engineering-team mentorship.

WORK EXPERIENCE

Trail Appliances • Richmond, BC • 10/2018 - 03/2026

Full Stack Software Developer • Full-time

- **Led an enterprise re-platforming** from a high-traffic Magento 2 monolith to a Commercetools microservices architecture, improving scalability, deployment flexibility, and long-term maintainability.
- **Served as the hands-on technical lead for a five-developer team**, spending approximately 70% of time on architecture and implementation and 30% on technical planning, code reviews, mentoring, defect resolution, and release readiness.
- **Built an AI shopping assistant (RAG)** using LangChain and LangGraph to deliver product Q&A, side-by-side comparisons, and personalized suggested questions, grounding responses in GraphQL product data and Cloudflare KV knowledge, with LangSmith tracing and graceful evaluation to guide quality improvements.
- **Implemented an event-driven product indexing pipeline** for approximately 8,700 products, using Node.js to publish product-change events from commercetools Subscriptions to Google Cloud Pub/Sub and Laravel with Mongodb on Cloud Run to apply delta updates to Algolia, supplemented by daily batched full syncs and automated retries.
- **Reduced cart and checkout response time by 67%** from approximately 3 seconds to 1 second by consolidating fragmented GraphQL calls and redesigning shared checkout state.
- **Delivered multiple payment integrations** across a React PWA and Magento 2 system, covering payment initialization, callback validation, mobile redirects, status polling, duplicate-safe order creation, and transaction-state handling.
- **Delivered full-stack commerce features** for catalog, search, cart, checkout, payments, accounts, and orders using React, Vue.js, Laravel, GraphQL, REST, and Commercetools.
- **Automated clearance inventory operations** with an n8n workflow that refreshed warehouse inventory hourly despite no direct API integration, saving approximately 20 staff hours per week.
- **Applied AI-assisted development (spec-driven)** to accelerate implementation, code exploration, testing, and debugging while validating output quality.
- **Improved production reliability** through New Relic monitoring, distributed tracing, and Slack alerts for checkout and payment failures, enabling faster root-cause analysis and incident resolution.
- **Established engineering practices** covering Git workflows, Azure DevOps pipelines, peer code reviews, QA validation, sprint planning, backlog management, and release governance.

Alive Health Centre • Richmond, BC • 07/2016 - 02/2018

Web Developer • Full-time

- **Developed and maintained a Magento 2 e-commerce platform** using PHP, JavaScript, MySQL, and Zend Framework, delivering platform enhancements, content migrations, and customer-experience improvements in partnership with business stakeholders.

Various Companies • 01/2013 - 01/2016

Web Developer and IT roles

- **Delivered web applications and digital solutions** across e-commerce, agency, travel, and higher-education environments using PHP, WordPress, Magento, Drupal, and CMS.
- **Built foundational experience in full-stack development**, systems integration, website operations, client communication, and production support.

EDUCATION

Bachelor of Computer Science

Vancouver Island University • Nanaimo, BC • 09/2011 - 05/2015

CERTIFICATIONS

AWS Certified Developer Associate • 01/2024

Amazon Web Services

Project Management Certificate Program • 01/2024

UBC Sauder School of Business

PROJECTS

AI Learning & Interview Platform • 06/2026 - 12/2026

- **Full-Stack & Real-Time Features:** Built a Vue 3/TypeScript and Python/FastAPI platform with MySQL persistence, token-based authentication, and SSE streaming; integrated RTC voice interviews with backend-issued access tokens, live captions, and persisted interview transcripts.
- **Multi-Agent Workflows & RAG:** Orchestrated assessment workflows with LangChain/LangGraph; built document ingestion with OCR fallback, chunking(Qdrant), and embeddings, enabling conditional, user-isolated retrieval with relevance filtering and source references linked to originals stored in MySQL.
- **Memory, Security & Resilience:** Implemented on-demand loading of recent dialogue, session summaries, and cross-session profiles to manage context length and personalize responses; added API rate limiting, password hashing, upload validation, and degradation for retrieval and cache failures.

SKILLS

Frontend Development: CSS, HTML, JavaScript, React, TypeScript, Vue

Backend & APIs: Fastapi, GraphQL, Java, Laravel, Node.js, Object-Oriented Programming, PHP, Python, REST APIs, Spring Boot, Webhooks

Architecture: Caching, Event-Driven Systems, Microservices, Software Architecture, System Integration

AI & Automation: LangChain, LangGraph, LangSmith, n8n, Qdrant, RAG, Vector Retrieval

Data & Messaging: Elasticsearch, Google Cloud Pub/Sub, MariaDB, MongoDB, MySQL, RabbitMQ, Redis

Cloud & DevOps: Application Monitoring, AWS, Azure DevOps, CI/CD, Docker, Git, GitHub, Linux, New Relic